Quantum Field Theory IV Week 5 Homework

Exercise 5.7

Exercise 5.7. *Use equivalence theorem to compute $W_L^+W_L^- \to W_L^+W_L^-$ scattering amplitude in the limit $\sqrt{s} \gg m_W$. Full calculation in the Standard Model would give*

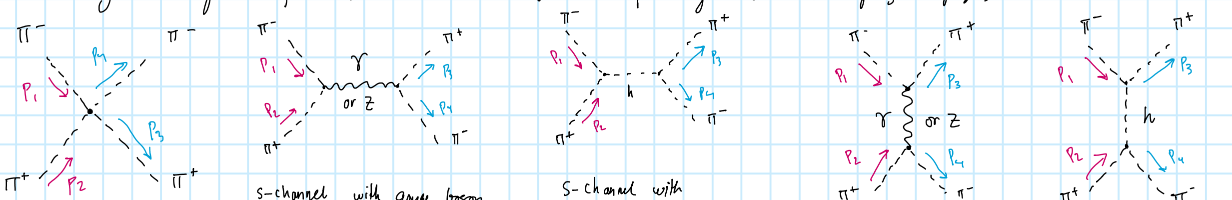
$$\mathcal{M}(W_L^+ + W_L^- \to W_L^+ + W_L^-) = \frac{g^2}{2}\left(\frac{m_H^2}{m_W^2} + \frac{s^2+st+t^2}{st\cos^2\theta_W}\right)$$

and you should obtain this by considering the scattering of the charged would-be-Goldstone bosons. Remember that $\lambda = m_H^2/(2v^2)$, $m_W = gv/2$ and $e = g\sin\theta_W$. There are seven diagrams you need to compute and add up.

The GB equivalence theorem states:

$$M(W_L^+, W_L^-, Z^\circ) = M(\pi^+, \pi^-, \pi^\circ)\left(1 + \mathcal{O}\left(\frac{m_W}{\sqrt{s}}\right)\right)$$

So to calculate $W^+W^- \to W^+W^-$ at high energies ($\sqrt{s} \gg m_W$) we need just calculate $\pi^+\pi^- \to \pi^+\pi^-$, where $\pi^\pm$ is a Goldstone boson.
The Feynman diagrams for $\pi^+\pi^- \to \pi^+\pi^-$ are following (we are using $\xi = 1$ gauge):



s-channel with gauge boson mediator

s-channel with higgs (scalar) mediator

t-channel with gauge boson mediator

t-channel with higgs mediator

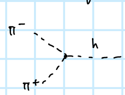
The set of momentum labels chosen along with the style of s-channel with the outgoing particles are presented help to make the whole calculation easier to visualize and consistent with the way t-channel works (no $\pi^-\pi^-(\gamma/Z/h)$-vertex)
Since the momentum labels are non-standard we must also give proper definition of Mandelstam variables:
Since $\sqrt{s}$ denotes the Centre of Mass (C.M.) energy it is straightforward to set $(P_1+P_2)^2 = s = (P_3+P_4)^2$
Since $\sqrt{t}$ denotes the momentum tranfer of the vertex, looking at the t-channel diagram we see that the momentum transfer is $P_1 - P_3$, we get the Mandelstam t to be $t = (P_1-P_3)^2 = (P_4-P_2)^2$
And finally the Mandelstam u is then defined as $u = (P_1-P_4)^2 = (P_3-P_2)^2$.
We also note the usefull relation $s+t+u = 4m_W^2$, since mass of $\pi^\pm$ is m_W in $\xi = 1$ gauge that we are using.
Since we are at high energy limit ($\sqrt{s} \gg m_W \Rightarrow E^2 \simeq |\vec{p}|^2 \Rightarrow p^2 = 0$), we can ignore the masses of $\pi^\pm$ and get:

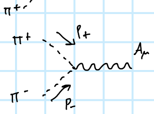
$$s \simeq 2P_1\cdot P_2 = 2P_3\cdot P_4 \quad , \quad t \simeq -2P_1\cdot P_3 = -2P_2\cdot P_4 \quad , \quad u \simeq -2P_1\cdot P_4 = -2P_2\cdot P_3 \quad \text{and} \quad u+s+t \simeq 0$$

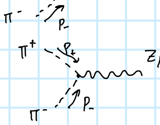
We use the Feynman rules from paper by Romão (Arxiv 1209.6213)

For 4-point vertex we get:  $= -\frac{i}{2}g^2\frac{m_h^2}{m_W^2}$

I'm writing higgs mass as M_h and not M_H

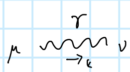
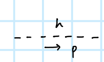
For Goldstone-Higgs 3-point vertex we get:  $= -\frac{i}{2}g\frac{m_h^2}{m_W}$

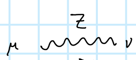
For Goldstone-photon 3-point vertex we get:  $= -i\eta_e e(p_+ - p_-)_\mu$

And lastly for Goldstone-Z-boson 3-point vertex we get:  $= -i\eta\eta_Z g\frac{\cos(2\theta_W)}{2\cos(\theta_W)}(p_+ - p_-)_\mu$

Here η, η_e, η_Z depend on sign convention and we will use $\eta = -1 = \eta_e$ and $\eta_Z = 1$ since it is the one used in Peskin & Schroeder.

In Feynman-'t Hooft gauge ($\xi = 1$) we get the following propagators:

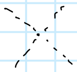
 $= -i\frac{g_{\mu\nu}}{k^2+i\varepsilon}$,  $= \frac{i}{p^2 - m_h^2 + i\varepsilon}$

 $= -i\frac{g_{\mu\nu}}{k^2 - m_Z^2 + i\varepsilon}$

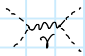
Now that we have the Feynman rules, we can calculate the amplitude $\mathcal{M}(\pi^+\pi^- \to \pi^+\pi^-)$

$$-i\mathcal{M} = \sum (\text{Feynman diagrams}) = -i \sum \mathcal{M}_{\text{diagrams}}$$

Let's write $\mathcal{M}$ for each of the 7 Feynman diagrams and in the end sum them over.



$: \quad -i\mathcal{M}_{fp} = -\frac{i}{2} g^2 \frac{m_h^2}{m_W^2}$, since the external legs of scalars give 1.



$$: \quad -i\mathcal{M}_{s\gamma} = -i\eta_e e \,(P_2 - P_1)_\mu \left[\frac{-ig^{\mu\nu}}{(P_1+P_2)^2}\right] \left[-i\eta_e e\,((-P_3)-(-P_4))_\nu\right]$$

$$= i\frac{e^2}{s} (P_2 - P_1)_\mu (P_4 - P_3)^\mu = i\frac{e^2}{s}\left[P_2\cdot P_4 - P_2\cdot P_3 - P_1\cdot P_4 + P_1\cdot P_3\right]$$

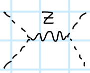
$$-i\mathcal{M}_{s\gamma} = i\frac{e^2}{s}\left[-t/2 + u/2 + u/2 - t/2\right] = -i\frac{e^2}{s}\left[t-u\right]$$

$\eta_e^2 = 1$

$(P_1+P_2)^2 = s$, where s is the

Use $P_2\cdot P_4 = -t/2 = P_1\cdot P_3$

$P_1\cdot P_4 = -u/2 = P_2\cdot P_3$



$$: \quad -i\mathcal{M}_{sZ} = -i\eta\eta_Z g \frac{\cos(2\theta_W)}{2\cos(\theta_W)} (P_2 - P_1)_\mu \left[-\frac{ig^{\mu\nu}}{(P_1+P_2)^2 - m_Z^2}\right]\left(-i\eta\eta_Z g \frac{\cos(2\theta_W)}{2\cos(\theta_W)}\right)(P_3 - P_4)_\nu$$

$$= \frac{ig^2}{s}\frac{\cos^2(2\theta_W)}{4\cos^2(\theta_W)} (P_2 - P_1)_\mu (P_3 - P_4)^\mu$$

From previous diagram we got $(P_2-P_1)_\mu (P_3-P_4)^\mu = -[t-u]$

$$-i\mathcal{M}_{sZ} = -\frac{ig^2}{s}\frac{\cos^2(2\theta_W)}{4\cos^2(\theta_W)}\left[t-u\right]$$

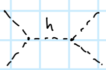
$\eta_Z^2 = 1 = \eta^2$

$(m_W = 80$ GeV & $m_Z = 91$ GeV$)$

Since $\sqrt{s} \gg m_W$

$\Rightarrow (P_1+P_2)^2 = s \gg m_W^2 \Rightarrow s \gg M_Z^2$

So we can ignore m_Z^2 in the propagator

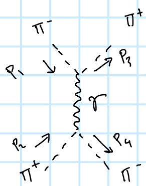


$$: \quad -i\mathcal{M}_{sh} = -\frac{i}{2} g \frac{m_h^2}{m_W}\left[\frac{i}{s - m_h^2}\right]\left(-\frac{i}{2} g \frac{m_h^2}{m_W}\right)$$

$$-i\mathcal{M}_{sh} = -\frac{i}{4s} g^2 \frac{m_h^4}{m_W^2}$$

Since we had $s \gg m_Z^2$ it is also reasonable to expect that we are looking at case where $s \gg m_h^2$ since $m_h \doteq 125$ GeV and $m_Z \simeq 91$ GeV, so they are basically of same order of magnitude

Now let's calculate $\mathcal{M}$ for t-channel processes:



$$: \quad -i\mathcal{M}_{t\gamma} = -i\eta_e e\,(-P_3 - P_1)_\mu \frac{-ig^{\mu\nu}}{(P_1-P_3)^2} (-i\eta_e e)(P_2 - (-P_4))_\nu$$

$$= -i\frac{e^2}{t} (P_3 + P_1)_\mu (P_2 + P_4)^\mu = -i\frac{e^2}{t}\left[P_3\cdot P_2 + P_3\cdot P_4 + P_1\cdot P_2 + P_1\cdot P_4\right]$$

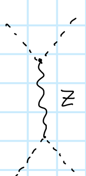
$$= -i\frac{e^2}{t}\left[-u/2 + \frac{s}{2} + \frac{s}{2} - u/2\right] = -i\frac{e^2}{t}\left[s-u\right]$$

$\eta_e^2 = 1$, $(P_1-P_3)^2 = t$

$P_1\cdot P_2 = s/2 = P_3\cdot P_4$

$P_2\cdot P_3 = -u/2 = P_1\cdot P_4$

$P_2\cdot P_4 = -t/2 = P_1\cdot P_3$



$$: \quad -i\mathcal{M}_{tZ} = -i\eta\eta_Z g \frac{\cos(2\theta_W)}{2\cos(\theta_W)} (-P_3 - P_1)_\mu \frac{-ig^{\mu\nu}}{t} (-i\eta\eta_Z g)\frac{\cos(2\theta_W)}{2\cos(\theta_W)} (P_2 - (-P_4))_\nu$$

$$= -i\frac{g^2}{t}\frac{\cos^2(2\theta_W)}{4\cos^2(\theta_W)} (P_3+P_1)_\mu (P_2+P_4)^\mu = -i\frac{g^2}{t}\frac{\cos^2(2\theta_W)}{4\cos^2(\theta_W)}\left[s-u\right]$$

$\eta^2 = \eta_Z^2 = 1$

we've already supressed the mass term from the propagator

$$: \quad -i\mathcal{M}_{th} = \left(-\frac{i}{2} g \frac{m_h^2}{m_W}\right)^2 \frac{i}{t} = -\frac{i}{4t} g^2 \frac{m_h^4}{m_W^2}$$

Now that we have amplitudes $\mathcal{M}$ for all 7 diagrams we can now sum them:

$$\Rightarrow -i\mathcal{M} = -i\mathcal{M}_{fp} - i\mathcal{M}_{s\gamma} - i\mathcal{M}_{sz} - i\mathcal{M}_{sh} - i\mathcal{M}_{t\gamma} - i\mathcal{M}_{tz} - i\mathcal{M}_{th}$$

$$\Rightarrow -i\mathcal{M} = -\frac{i}{2}g^2\frac{m_h^2}{m_w^2} - i\frac{e^2}{s}[t-u] - i\frac{g^2}{s}\frac{\cos^2(2\theta_w)}{4\cos^2(\theta_w)}[t-u] - \frac{i}{4s}g^2\frac{m_h^4}{m_w^2} - i\frac{e^2}{t}[s-u] - i\frac{g^2}{t}\frac{\cos^2(2\theta_w)}{4\cos^2(\theta_w)}[s-u] - \frac{i}{4t}g^2\frac{m_h^4}{m_w^2}$$

$$-i\mathcal{M} = -\frac{i}{2}g^2\frac{m_h^2}{m_w^2} - ie^2\left[\frac{t-u}{s} + \frac{s-u}{t}\right] - ig^2\frac{\cos^2(2\theta_w)}{4\cos^2(\theta_w)}\left[\frac{t-u}{s} - \frac{s-u}{t}\right] - \frac{i}{4}g^2\frac{m_h^2}{m_w^2}\left[\frac{1}{s}+\frac{1}{t}\right]$$

$e = g\sin\theta_w \Rightarrow e^2 = g^2\sin^2\theta_w$

$\cos(2\theta) = \cos^2\theta - \sin^2\theta$

$$\Rightarrow \frac{(\cos^2\theta - \sin^2\theta)^2}{\cos^2\theta} = \frac{\cos^4\theta + \sin^4\theta - 2\cos^2\theta\sin^2\theta}{\cos^2\theta}$$

$$= \frac{(\cos^2\theta + \sin^2\theta)^2 - 4\cos^2\theta\sin^2\theta}{\cos^2\theta}$$

$$= \frac{1}{\cos^2\theta} - 4\sin^2\theta$$

Let's first simplify $\frac{t-u}{s} + \frac{s-u}{t} = \frac{t^2 - u(t+s) + s^2}{st}$

$u = -s-t = -(s+t)$

$$\Rightarrow \frac{t-u}{s} + \frac{s-u}{t} = \frac{t^2 + (s+t)^2 + s^2}{st} = 2\,\frac{t^2+s^2+st}{st}$$

And now let's also use $e^2 = g^2\sin^2\theta_w$ and $\frac{\cos^2(2\theta_w)}{\cos^2\theta_w} = \frac{1}{\cos^2\theta_w} - 4\sin^2\theta_w$

$$\Rightarrow -i\mathcal{M} = -\frac{i}{2}g^2\frac{m_h^2}{m_w^2} - i\cancel{g^2\sin^2\theta_w}\left[2\,\frac{t^2+s^2+st}{st}\right] - ig^2\left[\frac{1}{4\cos^2\theta_w} - \cancel{\sin^2\theta_w}\right]\left[2\,\frac{t^2+s^2+st}{st}\right] - \frac{i}{4}g^2\frac{m_h^2}{m_w^2}\left[\frac{1}{s}+\frac{1}{t}\right]$$

$$\Rightarrow -i\mathcal{M} = -\frac{ig^2}{2}\frac{m_h^2}{m_w^2} - \frac{i}{2}g^2\,\frac{t^2+s^2+st}{st\cos^2\theta_w} - \frac{i}{4}g^2\frac{m_h^2}{m_w^2}\left[\frac{m_h^2}{s}+\frac{m_h^2}{t}\right]$$

, here we can set the last term to go to zero since $s \gg m_h^2$ & $t \gg m_h^2$

$$-i\mathcal{M} \underset{\uparrow}{=} -\frac{i}{2}g^2\frac{m_h^2}{m_w^2} - \frac{i}{2}g^2\,\frac{t^2+s^2+st}{st\cos^2\theta_w}$$

at high energy limit

$$\Rightarrow \mathcal{M} = \frac{g^2}{2}\left[\frac{m_h^2}{m_w^2} + \frac{s^2+st+t^2}{st\cos^2\theta_w}\right]$$

, where $\mathcal{M} = \mathcal{M}(\pi^+\pi^- \to \pi^+\pi^-)$

Since at high energy limit $\mathcal{M}(W^+W^- \to W_L^+W_L^-) = \mathcal{M}(\pi^+\pi^- \to \pi^+\pi^-)$, we get

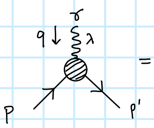
$$\mathcal{M}(W_L^+W_L^- \to W_L^+W_L^-) = \frac{g^2}{2}\left(\frac{m_h^2}{m_w^2} + \frac{s^2+st+t^2}{st\cos^2\theta_w}\right) \quad \square$$

Exercise 5.8

Exercise 5.8. *(4p.) Consider the contribution to muon electromagnetic vertex involving a W-neutrino loop diagram. Compute these contributions in the Feynman-'t Hooft gauge and extract the contribution to the anomalous magnetic moment of the muon. Note that you can take the limit $m_W^2 \gg m_\mu$ and also that for each diagram there is an accompanying diagram with the W replaced by Goldstone bosons. You should find that these diagrams yield a contribution*

$$a_\mu(\nu) = \frac{G_F m_\mu^2}{8\pi^2\sqrt{2}}\frac{10}{3}.$$

The muon magnetic moment is given by the following diagram:

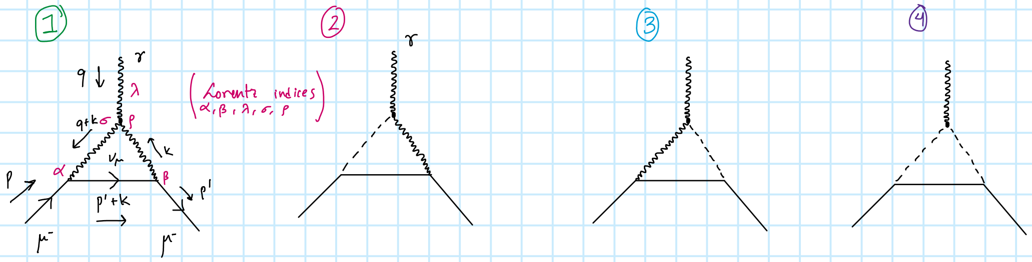


$= \bar{u}(p')\, ie\Gamma^\lambda(p,p')\, u(p)$, where $ie\Gamma^\lambda = ie\left(\gamma^\lambda F_1(q^2) + \frac{i}{2m}\sigma^{\lambda\rho} q_\beta\, F_2(q^2)\right)$

We are interested in finding the contribution to $F_2(q^2)$ so we may ignore/drop the terms that contribute to $F_1(q^2)$. All of the divergences that appear are in $F_1(q^2)$ term so we can set $D=4$ for dimensional integration from the start.
↳ Same for Dirac algebra stuff

We are using Feynman-'t Hooft gauge: $\xi = 1$ ⇒ The mass of the Goldstone boson is then: $M_G = \xi m_W = m_W$

In our gauge, we must consider the following Feynman diagrams:

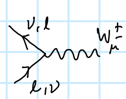


The momenta & Lorentz indices defined above are same for all diagrams (except when there is no need for Lorentz index for scalar propagators)

The result should be same for $\mu^-, \nu_\mu \to \mu^+\, \bar{\nu}_\mu$, but we do not care about that since we are looking for loop with neutrino and not anti-neutrino. As such it is not important label the name of the particles

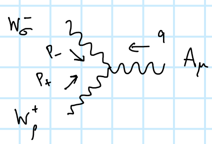
First let's write the Feynman rules (Once again from the paper by Romaŏ (Arxiv 1209.6213))
The convention we use has $\eta = \eta_e = -1$, $\xi_W = 1$, $P_L = \frac{1}{2}(1-\gamma_5)$, $P_R = \frac{1}{2}(1+\gamma_5)$

The lepton - W-boson vertex has the following vertex rule:



$= -i\eta \frac{g}{\sqrt{2}}\gamma_\mu P_L = i\frac{g}{\sqrt{2}}\gamma_\mu P_L$

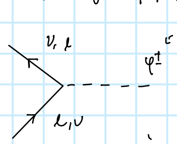
The Gauge boson 3-vertex for W-bosons and a photon gives us



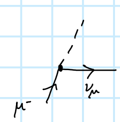
$$= -i\eta_e e\left[g_{\sigma\rho}(P_- - P_+)_\mu + g_{\rho\mu}(P_+ - q)_\sigma + g_{\mu\sigma}(q - P_-)_\rho\right]$$

$$= ie\left[g_{\sigma\rho}(P_- - P_+)_\mu + g_{\rho\mu}(P_+ - q)_\sigma + g_{\mu\sigma}(q - P_-)_\rho\right]$$

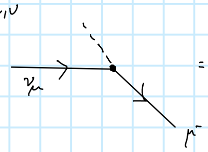
The lepton - Goldstone boson vertex gives:



$= -i\frac{g}{\sqrt{2}}\frac{m_\ell}{m_W}P_{R,L}$, which give us:



$= -i\frac{g}{\sqrt{2}}\frac{m_\mu}{m_W}P_R$ and



$= -i\frac{g}{\sqrt{2}}\frac{m_\mu}{m_W}P_L$

There are two relevant Goldstone boson - gauge boson vertices and they give following rules.

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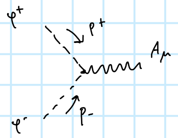
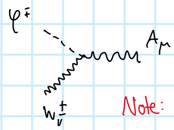


$= -i\eta_e\, e(p_- - p_-)_\mu = ie(p_+ - p_-)_\mu$

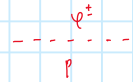


$= i\eta_e e m_w g_{\mu\nu} = iem_w g_{\mu\nu}$!!!

Note: this is actually wrong Feynman-rule since it leads to $\frac{7}{3} - 1 = \frac{4}{3}$ at the end instead of $\frac{7}{3} + 1 = \frac{10}{3}$ with $-iem_W g_{\mu\nu}$.

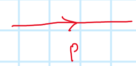
$\Rightarrow$ I'll be using $-iem_W g_{\mu\nu}$, since it leads to correct result.

We also have the following propagators:



$= \dfrac{i}{p^2 - \xi_W m_W^2 + i\varepsilon} = \dfrac{i}{p^2 - m_W^2 + i\varepsilon}$ and

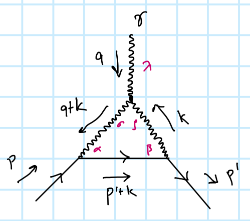
$= \dfrac{-i\, g^{\mu\nu}}{k^2 - m_W^2 + i\varepsilon}$



$= \dfrac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon}$, since we have neutrino propagator we may set $m = 0$ since neutrino has low mass.

We see that in the gauge we are using, the propagators of W-bosons and Goldstone bosons are very similar with the denominator being the same. This means that every diagram has same denominator and as such calculating the denominator of the first diagram in terms of Feynman Feynman parameters means doing same for all all of diagrams as such we can integrate over momenta of all diagrams under same integral since all will have same variable l to integrate over. This also instructs us to sum the numerators before integration.

Let's start with the diagram 1.



$$= \bar{u}(p')\int \frac{d^4k}{(2\pi)^4}\left[i\frac{g}{\sqrt{2}}\gamma^\beta P_L\right]\frac{i(\not{p}'+\not{k})}{(p'+k)^2 + i\varepsilon}\left[i\frac{g}{\sqrt{2}}\gamma^\alpha P_L\right] ie\left[g^{\sigma\rho}(q+k+k)^\lambda + g^{\rho\lambda}(q-k)^\sigma - g^{\lambda\sigma}(q+q+k)^\rho\right]$$

$$\times \frac{-i\, g_{\sigma\alpha}}{(q+k)^2 - m_W^2 + i\varepsilon}\;\frac{-i\, g_{\rho\beta}}{k^2 - m_W^2 + i\varepsilon} \equiv I_1$$

// We suppress $i\varepsilon$ from here on

$$I_1 = -e\frac{g^2}{2}\int\frac{d^4k}{(2\pi)^4}\left[g^{\sigma\rho}(q+2k)^\lambda + g^{\rho\lambda}(q-k)^\sigma + g^{\lambda\sigma}(-2q-k)^\rho\right]\frac{g_{\rho\beta}\,g_{\sigma\alpha}}{4}\;\frac{\bar{u}(p')\left(\gamma^\beta(1-\gamma^5)(\not{p}'+\not{k})\gamma^\alpha(1-\gamma^5)\right)u(p)}{\left((q+k)^2 - m_W^2\right)\left(k^2 - m_W^2\right)(p'+k)^2}$$

$$I_1 \equiv e\frac{g^2}{4}\int\frac{d^4k}{(2\pi)^4}\frac{N_1^\lambda}{D}$$

Let's start by writing this in terms of Feynman Parameters:

$$\frac{1}{ABC} = \int_0^1 dx\,dy\,dz\; 2\,\frac{\delta(x+y+z-1)}{(xA + yB + zC)^3} = \int dx\,dy\,\frac{2}{[xA + yB + (1-x-y)C]^3}$$

$\Rightarrow$

$$I_1 = e\frac{g^2}{4}\int dx\,dy\,dz\; 2\delta(x+y+z-1)\;\frac{N_1^\lambda}{2\left[x\left((q+k)^2 - m_W^2\right) + y\left(k^2 - m_W^2\right) + z(p'+k)^2\right]^3}$$

Let's simplify the following term inside the denominator: $D' = x\left((q+k)^2 - m_W^2\right) + y\left(k^2 - m_W^2\right) + zk^2$

$$\Rightarrow\; D' = x\left[q^2 + k^2 + 2qk - m_W^2\right] + y\left[k^2 - m_W^2\right] + z\left[k^2 + 2p'k + p'^2\right] \qquad // \; x+y+z = 1$$

$$= k^2 + 2\left[xq + zp'\right]k + xq^2 + zp'^2 - m_W^2(x+y)$$

// We do a shift $l = k + xq + zp'$ such that $d^4k \to d^4l$

$$= l^2 - \left[xq + zp'\right]^2 + xq^2 + zp'^2 - m_W^2(x+y)$$

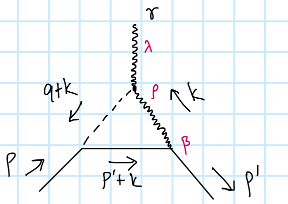
$$D' \equiv \ell^2 - \Delta \quad , \quad \text{with} \quad \Delta = (xq + zp')^2 - xq^2 - zp'^2 + m_W^2 (x+y)$$

While we could simplify Δ with approximations here, we leave that for the end of the calculation

$$\Rightarrow \quad I_1 = e\frac{g^2}{4} \int dx\, dy\, dz\, \delta(x+y+z-1) \int \frac{d^4\ell}{(2\pi)^4} \frac{N_1^\lambda}{(\ell^2 - \Delta)^3}$$

Before continuing and simplifying N_1^λ, we will first collect the Numerators of Integrals of diagram ② and ③. We will neglect diagram ④ since it has additional m_W^{-1} multiplying it because of Goldstone-muon-vertex which is proportional to m_μ / m_W. The other diagrams don't have this problem since Goldstone-W-photon-vertex contributes m_W to cancel the single $\frac{1}{m_W}$ arising from Goldstone-muon-vertex. So since $m_\mu \ll m_W$ we have ④ $\ll$ ① + ② + ③.

Now let's find N_2^λ from diagram ②:



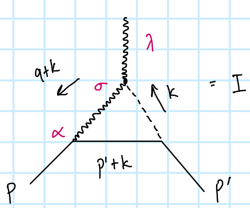
from 3-vertex, I'm doing the correction to the sign of one of the rules with this colour to show how it impact calculation and why it's necessary.

$$= \bar{u}(p') \int \frac{d^4k}{(2\pi)^4} \left[i \frac{g}{\sqrt{2}} \gamma^\beta P_L \right] \frac{i(\not{p}' + \not{k})}{(p'+k)^2 + i\varepsilon} \left[-i \frac{g}{\sqrt{2}} \frac{m_\mu}{m_W} P_R \right] \underbrace{\frac{i(-ie m_W g^{\rho\lambda})}{(q+k)^2 - m_W^2 + i\varepsilon}}_{\text{Goldstone propagator}} \underbrace{\frac{-i g_{\rho\beta}}{k^2 - m_W^2 + i\varepsilon}}_{\text{W- propagator}}$$

$$= +e \frac{g^2}{4} m_\mu \int \frac{d^4k}{(2\pi)^4} \frac{1}{2} \frac{\bar{u}(p')[\gamma^\lambda (1-\gamma^5)(\not{p}' + \not{k})(1+\gamma^5)] u(p)}{\underbrace{((q+k)^2 - m_W^2)(k^2 - m_W^2)(p'+k)^2}_{\text{Same denominator as for } I_1\text{-integral so same Feyman parameters.}}} \equiv I_2$$

$$\Rightarrow \quad I_2 = e\frac{g^2}{4} \int dx dy dz\, \delta(x+y+z-1) \int \frac{d^4\ell}{(2\pi)^4} \frac{N_2^\lambda}{(\ell^2 - \Delta)^3} \quad , \quad \text{with} \quad N_2^\lambda = +m_\mu\, \bar{u}(p') [\gamma^\lambda (1-\gamma^5)(\not{p}' + \not{k})(1+\gamma^5)] u(p)$$

Now let's find N_3^λ from diagram ③:



3-vertex

$$= I_3 \equiv \bar{u}(p') \int \frac{d^4k}{(2\pi)^4} \left[-i \frac{g}{\sqrt{2}} \frac{m_\mu}{m_W} P_L \right] \frac{i(\not{p}' + \not{k})}{(p'+k)^2} \left[i \frac{g}{\sqrt{2}} \gamma^\alpha P_L \right] \underbrace{\frac{-i g_{\alpha\sigma}}{(k+q)^2 - m_W^2}}_{\text{W- propagator}} \underbrace{\frac{i(-ie m_W g^{\lambda\sigma})}{k^2 - m_W^2}}_{\text{Goldstone-propagator}}$$

$$\Rightarrow \quad I_3 = e \frac{g^2}{4} \int \frac{d^4k}{(2\pi)^4} \frac{+m_\mu \bar{u}(p')[(1-\gamma^5)(\not{p}' + \not{k})(1-\gamma^5)\gamma^\lambda] u(p)}{\underbrace{2((q+k)^2 - m_W^2)(k^2 - m_W^2)(p'+k)^2}_{\text{Again the same denominator that leads to same Feyman parameters}}}$$

$$I_3 = e \frac{g^2}{2} \int dx dy dz\, \delta(x+y+z-1) \int \frac{d^4\ell}{(2\pi)^4} \frac{N_3^\lambda}{(\ell^2 - \Delta)^3} \quad , \quad \text{with} \quad N_3^\lambda = +m_\mu \bar{u}(p') [(1-\gamma^5)(\not{p}' + \not{k}) \gamma^\lambda (1-\gamma^5)] u(p)$$

All in all, we have

$$I \equiv I_1 + I_2 + I_3 = e\frac{g^2}{4} \int dx dy dz\, \delta(x+y+z-1) \int \frac{d^4\ell}{(2\pi)^4} \frac{N^\lambda}{(\ell^2 - \Delta)^3} \quad , \quad \text{with} \quad N^\lambda = N_1^\lambda + N_2^\lambda + N_3^\lambda$$

Now let's compute N^λ to a desirable form, but first small detour: let's simplify N_1^λ a bit before we move on to N^λ:

Now let's compute N^λ to a desirable form, but first small detour: let's simplify N_1^λ a bit before we move on to N^λ:

$$N_1^\lambda = -\left[g^{\sigma\rho}(q+2k)^\lambda + g^{\rho\lambda}(q-k)^\sigma + g^{\lambda\sigma}(-2q-k)^\rho \right] \bar{u}(p')\left(\gamma_\rho(1-\gamma^5)(\not{p}'+\not{k})\gamma_\sigma(1-\gamma^5)\right)u(p)$$

Since electromagnetic form factors conserve parity we can neglect the terms linear in γ^5 (must include $(\gamma^5)^2 = 1$ - terms, as I learned the hard way....) from the diagrams (parity conservation ensures that they would cancel at the end).
So we get:

$$\Rightarrow N_1^\lambda = -\left[g^{\sigma\rho}(q+2k)^\lambda + g^{\rho\lambda}(q-k)^\sigma + g^{\lambda\sigma}(-2q-k)^\rho \right] \bar{u}(p')\left[\gamma_\rho(\not{p}'+\not{k})\gamma_\sigma + \gamma_\rho\gamma^5(\not{p}'+\not{k})\gamma_\sigma\gamma^5\right]u(p)$$

$$\Rightarrow N_1^\lambda = -\left[g^{\sigma\rho}(q+2k)^\lambda + g^{\rho\lambda}(q-k)^\sigma + g^{\lambda\sigma}(-2q-k)^\rho \right] \bar{u}(p')\left[\gamma_\rho(\not{p}'+\not{k})\gamma_\sigma + \gamma_\rho(\not{p}'+\not{k})\gamma_\sigma\right]u(p)$$

$$N_1^\lambda = -2\,\bar{u}(p')\left[\gamma^\rho(\not{p}'+\not{k})\gamma_\rho(q+2k)^\lambda + \gamma^\lambda(\not{p}'+\not{k})(\not{q}-\not{k}) - (2\not{q}+\not{k})(\not{p}'+\not{k})\gamma^\lambda\right]u(p)$$

To simplify this we use the following tools:

Dirac equation: $\bar{u}(p')\not{p}' = m_\mu \bar{u}(p')$, $\not{p}\,u(p) = m_\mu u(p)$ $\Rightarrow$ $\bar{u}(p')\not{q}\,u(p) = 0$

Dirac algebra: $\{\gamma^\alpha, \gamma^\beta\} = 2g^{\alpha\beta}$ $\Rightarrow$ $\not{a}\not{b} = 2a\cdot b - \not{b}\not{a}$ and numerator algebra from Peskin appendix A3

Now since the form factors appear in the vertex with following form,

 $= \bar{u}(p')\,ie\Gamma^\mu u(p)$, $ie\Gamma^\mu = ie\left(\gamma^\mu F_1(q^2) + \frac{i\sigma^{\alpha\beta}}{2m_\mu}q_\beta F_2(q^2)\right)$,

and we only want contribution to $F_2(q^2)$, we may drop from N_1^λ any and all terms with scalar $\cdot\,\gamma^\lambda$ - Dirac structure. All other terms must be considered to figure out $\frac{i\sigma^{\alpha\beta}}{2m_\mu}$ - contribution. We can also drop terms like q^μ due to Ward Identity: $q_\lambda\Gamma^\lambda = 0$

Let's simplify N_1^λ term by term:

$k = \ell - xq - zp'$

the terms with odd powers of ℓ vanish due to symmetric integration and $\not{\ell}\,\ell^\lambda = \gamma_\beta \ell^\beta \ell^\lambda$ can be dealt using the same symmetry: $\gamma_\beta \ell^\beta \ell^\lambda = \frac{1}{4}\ell^2 g^{\beta\lambda}\gamma_\beta = \frac{1}{4}\ell^2\gamma^\lambda$ and we can neglect this term since it has scalar $\cdot\,\gamma^\lambda$ -structure. This analysis works for all the $\not{\ell}$ - terms that appear in the N^λ so when we do $k = \ell - xq - x$ we can always drop the ℓ - term.

$$\bar{u}(p')\left[\gamma^\rho(\not{p}'+\not{k})\gamma_\rho(q+2k)^\lambda\right]u(p) = -2\,\bar{u}(p')\left[(\not{p}'+\not{k})(q+2k)^\lambda\right]u(p)$$

$$= -2\,\bar{u}(p')\left[(m_\mu - x\not{q} - z m_\mu)(q - 2xq - 2zp')^\lambda\right]u(p) \quad \| \ \bar{u}\not{q}u = 0, \ q^\lambda\text{-term} \Rightarrow 0$$

$$= +4\,\bar{u}(p')\left[z(1-z)m_\mu p'^\lambda\right]u(p), \quad \text{let's now use } 2p'^\lambda = \{\gamma^\lambda, \not{p}'\}$$

$$= 2\,\bar{u}(p')\left[z(1-z)m_\mu\left[\gamma^\lambda\not{p}' + \not{p}'\gamma^\lambda\right]\right]u(p) \quad \| \ \bar{u}(p')\not{p}' = m_\mu \Rightarrow m_\mu\gamma^\lambda \to \text{drop}$$

$$= 2\,\bar{u}(p')\left[z(1-z)m_\mu\gamma^\lambda(\not{q} + \underbrace{m_\mu}_{\text{drop}})\right]u(p)$$

$$= 2\,\bar{u}(p')\left[z(1-z)m_\mu\gamma^\lambda\not{q}\right]u(p)$$

Now let's move on to the 2nd term:

$$\bar{u}(p')\left[\gamma^\lambda(\not{p}'+\not{k})(\not{q}-\not{k})\right]u(p) \quad \| \ \ell = k - xq - zp' \text{ and drop } \ell$$

$$= \bar{u}(p')\left[\gamma^\lambda\left(\not{p}'(1-z) - x\not{q}\right)\left(\not{q}(1+x) + z\not{p}'\right)\right]u(p) \quad \| \ \not{p}' = \not{q} + \not{p} \text{ and } \not{p}\,u(p) = m_\mu u(p)$$

$$= \bar{u}(p)\left[\gamma^\lambda\left(\not{q}(\underbrace{1-z-x}_{=y}) + \not{p}(1-z)\right)\left(\not{q}(\underbrace{1+x+z}_{2-y}) + z m_\mu\right)\right]u(p) = \bar{u}(p')\left[\gamma^\lambda\left(\underbrace{q^2 y(2-y)}_{\text{drop}} + zy m_\mu\not{q} + (1-z)(2-y)\not{p}\not{q} + \underbrace{z(1-z)m_\mu^2}_{\text{drop}}\right]u(p)$$

$$= \bar{u}(p')\left[\gamma^\lambda\left(\not{q}\underbrace{(1-z-x)}_{=y} + \not{p}(1-z)\right)\left(\not{q}\underbrace{(1+x+z)}_{2-y} + z m_\mu\right)\right]u(p) = \bar{u}(p')\left[\gamma^\lambda\left(\underbrace{q^2 y(2-y)}_{\text{drop}} + zy m_\mu \not{q} + (1-z)(2-y)\not{p}\not{q} + \underbrace{z(1-z)m_\mu^2}_{\text{drop}}\right)\right]u(p)$$

$$= \bar{u}(p')\left[\gamma^\lambda \not{q}\, m_\mu zy + \gamma^\lambda(1-z)(2-y)\left[\underbrace{2p\cdot q}_{\text{drop}} - \not{q} m_\mu\right]\right]u(p) = \bar{u}(p')\left[\gamma^\lambda \not{q} m_\mu\left(zy - (1-z)(2-y)\right)\right]u(p)$$ || $zy - [2-y-2z+zy] = -2+y+2z$

$$= \bar{u}(p')\left[(y+2z-2)m_\mu \gamma^\lambda \not{q}\right]u(p)$$

Now let's do the final term:

$$-\bar{u}(p')\left[(2\not{q}+\not{k})(\not{p}'+\not{k})\gamma^\lambda\right]u(p)$$ || $k = l - xq - zp'$ and drop l

$$= -\bar{u}(p')\left[\left((2-x)\not{q} - z m_\mu\right)\left(\not{p}'(1-z) - x\not{q}\right)\gamma^\lambda\right]u(p) = -\bar{u}(p')\left[(2-x)(1-z)\not{q}\not{p}' - \underbrace{x(2-x)q^2}_{\text{drop}} - \underbrace{z(1-z)m_\mu^2}_{\text{drop}} + xz m_\mu \not{q}\right]\gamma^\lambda u(p)$$ || $\not{q}\not{p}' = 2q\cdot p - \not{p}\not{q}$

$$= -\bar{u}(p')\left[(2-x)(1-z)\left[\underbrace{2q\cdot p}_{\text{drop}} - m_\mu \not{q}\right] + xz m_\mu \not{q}\right]\gamma^\lambda u(p) = -\bar{u}(p')\left[-(2-x)(1-z) + xz\right]m_\mu \not{q}\gamma^\lambda u(p)$$

|| $xz - [2-2z-x+xz] = 2z+x-2$ || $x = 1-y-z$
$= 2z+1-y-z-2 = -1+z-y$

$$= -\bar{u}(p')\left[(-1-z+y)m_\mu \not{q}\gamma^\lambda\right]u(p)$$ || $\not{q}\gamma^\lambda = 2q^\lambda - \not{q}\gamma^\lambda$ and $q^\lambda \to 0$ due to Word Identity

$$= \bar{u}(p')\left[(-1+z-y)m_\mu \gamma^\lambda \not{q}\right]u(p)$$

In total our N_1^λ becomes:

$$N_1^\lambda = -2\bar{u}(p')\left[2z(1-z)m_\mu \gamma^\lambda \not{q} + (y+2z-2)m_\mu \gamma^\lambda \not{q} + (-1+z-y)m_\mu \gamma^\lambda \not{q}\right]u(p)$$

$$= -2\bar{u}(p')\left[2z(1-z) + (y+2z-2) + (-1+z-y)\right]m_\mu \gamma^\lambda \not{q}\, u(p)$$

|| $\gamma^\lambda \not{q} = \frac{1}{2}[\gamma^\lambda \not{q} + \gamma^\lambda \not{q}] = \frac{1}{2}[\gamma^\lambda, \gamma^\beta]q_\beta$ the last equality comes from Ward identity Since $\gamma^\lambda \not{q} = 2q^\lambda - \not{q}\gamma^\lambda$ and $q^\lambda \to 0$ due to Ward id.

$$= -\bar{u}(p')\left[2z(1-z) - 3 + 3z\right]m_\mu[\gamma^\lambda, \gamma^\beta]q_\beta\, u(p)$$

|| $2z(1-z) - 3 + 3z = 2z(1-z) - 3(1-z) = (2z-3)(1-z)$, also from definition of $\sigma^{\lambda\beta}$ we get $[\gamma^\lambda, \gamma^\beta] = -2i\sigma^{\lambda\beta}$

$$N_1^\lambda = -\bar{u}(p')\left[2(1-z)(3-2z)2m_\mu^2 \frac{i\sigma^{\lambda\beta}}{2m_\mu}q_\beta\right]u(p)$$

Now N_1^λ is in a desired form, let's now look at N_2^λ and N_3^λ, we start with N_3^λ:

$$N_3^\lambda = +m_\mu \bar{u}(p')\left[(1-\gamma^5)(\not{p}'+\not{k})\gamma^\lambda(1-\gamma^5)\right]u(p)$$, ignoring terms linear in γ^5 we get:

$$= +2m_\mu \bar{u}(p')\left[(\not{p}'+\not{k})\gamma^\lambda\right]u(p)$$ || $k = l - xq - zp'$ and drop l

$$= +2m_\mu \bar{u}(p')\left[(m_\mu - x\not{q} - z m_\mu)\gamma^\lambda\right]u(p)$$ || drop scalar $\cdot\,\gamma^\lambda$ terms

$$= -2m_\mu \bar{u}(p')\left[x\not{q}\gamma^\lambda\right]u(p)$$ || using Dirac algebra and Ward Idntity we can replace $\not{q}\gamma^\lambda$ with $-\gamma^\lambda\not{q}$

$$= +2m_\mu \bar{u}(p')\left(x\gamma^\lambda \not{q}\right)u(p)$$ || With same logic we replace $\gamma^\lambda\not{q}$ with $\frac{1}{2}[\gamma^\lambda, \gamma^\beta]q_\beta$, just as we did for N_1^λ also $\frac{1}{2}[\gamma^\lambda, \gamma^\beta] = -i\sigma^{\lambda\beta}$

$$= -2x m_\mu \bar{u}(p')\left[i\sigma^{\lambda\beta}q_\beta\right]u(p)$$

$$N_3^\lambda = -\bar{u}(p')\left[4x m_\mu^2\left(i\frac{\sigma^{\lambda\beta}}{2m_\mu}q_\beta\right)\right]u(p)$$, Now let's do same for N_2^λ:

$$\Rightarrow N_2^\lambda = +m_\mu \bar{u}(p')\left[\gamma^\lambda(1-\gamma^5)(\not{p}'+\not{k})(1+\gamma^5)\right]u(p)$$, ignoring terms linear in γ^5 we get:

$$= +2m_\mu \bar{u}(p')\left[\gamma^\lambda(\not{p}'+\not{k})\right]u(p)$$ || $k = l - xq - zp'$ and drop l

$$= +2m_\mu \bar{u}(p')\left[\gamma^\lambda\left(\not{p}'(1-z) - x\not{q}\right)\right]u(p)$$ || $\not{p}' = \not{q} + \not{p}$, $1-x = z+y$

$$N_2^\lambda = +2m_\mu \bar{u}(p')\left[\gamma^\lambda\left((1-z-x)\not{q} + (1-z)m_\mu\right)\right]u(p) = +2m_\mu \bar{u}(p')\left[\gamma^\lambda \not{q}\, y\right]u(p)$$ || from earlier: $\gamma^\lambda \not{q} \to \frac{1}{2}[\gamma^\lambda, \gamma^\beta]q_\beta$

$$= +2m_\mu \bar{u}(p')\left[\gamma^\lambda\left(\not{p}'(1-z) - x\not{q}\right)\right]u(p) \qquad \Big\| \quad \not{p}' = \not{q} + \not{p} \;, \quad 1-x = z+y$$

$$N_2^\lambda = +2m_\mu \bar{u}(p')\left[\gamma^\lambda\left(\underbrace{(1-z-x)}_{=y}\not{q} + (1-z)\underbrace{m_\mu}_{\text{drop}}\right)\right]u(p) = +2m_\mu\bar{u}(p')\left[\gamma^\lambda \not{q}\, y\right]u(p)$$

$\Big\|$ from earlier : $\gamma^\lambda \not{q} \to \frac{1}{2}[\gamma^\lambda, \gamma^\beta] q_\beta$

$\Rightarrow \frac{1}{2}[\gamma^\lambda, \gamma^\beta] = -i\sigma^{\lambda\beta}$

$$= -2m_\mu y\, \bar{u}(p')\left[2m_\mu \frac{i\sigma^{\lambda\beta}}{2m_\mu} q_\beta\right]u(p) = -\bar{u}(p')\left[4m_\mu^2\, y \left(\frac{i\sigma^{\lambda\beta}}{2m_\mu} q_\beta\right)\right]u(p)$$

Collecting these, we get :

$$\Rightarrow N^\lambda = N_1^\lambda + N_2^\lambda + N_3^\lambda = -\bar{u}(p')\left[2(1-z)(3-2z)2m_\mu^2 + 4m_\mu^2 y + 4m_\mu^2 x\right]\left(\frac{i\sigma^{\lambda\beta}}{2m_\mu}\right)u(p) \qquad \Big\| -(x+y) = -(1-z)$$

$$= -\bar{u}(p')\left[4m_\mu^2(1-z)(3-2z+1)\right]\left(\frac{i\sigma^{\lambda\beta}}{2m_\mu}\right)u(p)$$

$$N^\lambda = -\bar{u}(p)\left[8m_\mu^2(1-z)(2-z)\right]\left(\frac{i\sigma^{\lambda\beta}}{2m_\mu}\right)u(p)$$

|| this is where sign correction ends, which leads to correct result, if I had calculated all diagrams separately the $\frac{7}{3}-1$ vs $7/3+1$ issue would become more apparent, but unfortunately I noticed that late in the calculation so it's stay this way.

$\int_0^1 (1-z)(2-z) = \frac{5}{3}$

$\int_0^1 (1-z)^2 = \frac{1}{3}$

this is the difference

Now we are ready to do the final step of integrating everything :

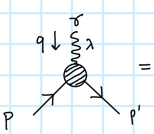
$$I = \bar{u}(p')\left\{-e\left[\frac{g^2}{4}\int dx\,dy\,dz\,\delta(x+y+z-1)\, 8m_\mu^2(1-z)(2-z)\left(\frac{i\sigma^{\lambda\beta}}{2m_\mu}q_\beta\right)\int\frac{d^4\ell}{(2\pi)^4}\frac{1}{(\ell^2-\Delta)^3}\right]\right\}u(p)$$

to evaluate the ℓ-integral we use eq. (A.44) from Peskin & Schroeder:

$$\int\frac{d^4\ell}{(2\pi)^4}\frac{1}{(\ell^2-\Delta)^3} = \frac{-i}{(4\pi)^2}\frac{\Gamma(1)}{\Gamma(3)}\frac{1}{\Delta} \qquad \Big\| \;\Gamma(3) = 2$$

$$\Rightarrow I = \bar{u}(p')\, ie\left[\frac{i\sigma^{\lambda\beta}}{2m_\mu} q_\beta \int dx\,dy\,dz\,\delta(x+y+z-1)\,\cancel{4}m_\mu^2(1-z)(2-z)\frac{g^2 m_\mu^2}{\cancel{4}\,(4\pi)^2}\frac{1}{\Delta}\right]u(p) \qquad \Big\| \;\Delta = \Delta(q^2)$$

Comparing with,



$$= \bar{u}(p')\, ie\Gamma^\lambda(p,p')\, u(p) \;, \text{ where } \quad ie\Gamma^\lambda = ie\left(\gamma^\lambda F_1(q^2) + \frac{i}{2m}\sigma^{\lambda\beta} q_\beta F_2(q^2)\right),$$

we see that we indeed have pure contribution to $F_2(q^2)$, the anomalous magnetic moment is $F_2(0)$

So we get the contribution of W-neutrino loop diagram to be

$$a_\mu(\nu) = \frac{g^2 m_\mu^2}{16\pi^2}\int dx\,dy\,dz\,\delta(x+y+z-1)(1-z)(2-z)\frac{1}{\Delta} \quad , \text{ where } \quad \Delta = \Delta(q^2=0)$$

The term Δ is given by $\Delta = (xq+zp')^2 - xq^2 - zp'^2 + m_W^2(x+y)$, we now want to approximate with $m_\mu \ll m_W$ but first we must find q^2 terms and set them to 0

$$\Rightarrow \Delta(q^2) = \left(x^2q^2 + z^2\underbrace{p'^2}_{m_\mu^2} + 2xz(p'-p)p'\right) - xq^2 - zm_\mu^2 + m_W^2(x+y)$$

In the $m_\mu \ll m_W$ limit we can set $m_\mu = 0$ for $\Delta(q^2)$ since m_W^2-term dominates over m_μ^2 ones

$$\Rightarrow \Delta(q^2) \approx \left(x^2q^2 - 2pp'\,xz\right) - xq^2 + m_W^2(x+y) \qquad \Big\| \; q^2 = -2p\cdot p' \text{ when } m_\mu \to 0$$

$$\Delta(q^2) \approx x^2q^2 - q^2xz - xq^2 + m_W^2(x+y) \qquad \Big\| \text{ now we set } q^2 = 0 \text{ for our } F_2(0) \text{ situation}$$

$$\Rightarrow \Delta(0) \approx m_W^2(x+y) = m_W^2(1-z)$$

$$\frac{g^2 m_\mu^2}{\ldots}\int^1 dx\,dy\,dz\;\delta(x+y+z-1)\frac{(1-z)(2-z)}{(1-z)} \qquad \Big\| \text{ first integrate over } y$$

$\Rightarrow \quad \Delta(0) \simeq m_W^2 (x+y) = m_W^2 (1-z)$

$$\Rightarrow \quad a_\mu(\nu) = \frac{g^2 m_\mu^2}{16\pi^2 m_W^2} \int_0^1 dx\, dy\, dz \;\; \delta(x+y+z-1) \; \frac{(1-z)(2-z)}{(1-z)}$$ || first integrate over y

$$= \frac{g^2 m_\mu^2}{16\pi^2 m_W^2} \int_0^1 dz \int_0^{1-z} dx \;\; (2-z)$$

$$= \frac{g^2 m_\mu^2}{16\pi^2 m_W^2} \int_0^1 dz \;\; (1-z)(2-z)$$ || $G_F = \frac{g^2}{8 m_W^2} \sqrt{2} \Rightarrow \frac{G_F}{\sqrt{2}} = \frac{g^2}{8 m_W^2}$

Simple integral, but we use calculator to calculate it

$$a_\mu(\nu) = \frac{G_F m_\mu^2}{2\pi^2 \sqrt{2}} \left[\frac{5}{6} \right] = \frac{G_F m_\mu^2}{8\pi^2 \sqrt{2}} \, \frac{10}{3} \quad \square$$

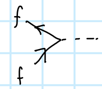
Exercise 5.9

Exercise 5.9. *Another contribution arises from a Z-muon loop diagram. Compute these in the Feynman-'t Hooft gauge so that you can extract the contribution to the anomalous magnetic moment. You should find that this contribution is*

$$a_\mu(Z) = -\frac{G_F m_\mu^2}{8\pi^2\sqrt{2}}\frac{1}{3}\left(4 + 8\sin^2\theta_W - 16\sin^4\theta_W\right).$$

We have two Feynman diagrams to consider

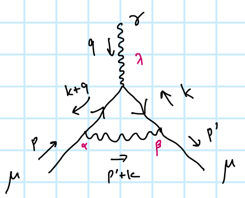
, the second loop is small compared to first one by order of m_μ^2/m_Z^2 since the following vertex the following form:



$= -g T_f^3 \frac{m_f}{m_W}\gamma_5$ (f = fermion)

$\Rightarrow$ and $m_\mu \ll m_Z$ so this is very small compared to first one and as such we ignore it.

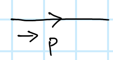
So we only have to calculate the following Feynman diagram



Lorentz indices : λ, α, β

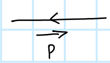
, We have the following Feynman rules (from Arxiv 1209.6213) :

$= -ie\gamma^\alpha$

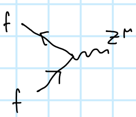


$= \frac{i(\not{p} + m_\mu)}{p^2 - m_f^2 + i\varepsilon}$

$\Rightarrow$

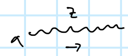


$= \frac{i(-\not{p} + m_\mu)}{p^2 - m_f^2 + i\varepsilon}$



$= i\frac{g}{\cos\theta_W}\gamma^\mu\left(g_V^f - g_A^f\gamma_5\right) \equiv i\frac{g}{\cos\theta_W}\gamma^\mu g_{VA}$

The Z-propagator in the gauge $\xi = 1$ is



$= -i\frac{g_{\alpha\beta}}{k^2 - m_W^2 + i\varepsilon}$

Now that we have the Feynman rules we can write the expression for the diagram:

$$= \int \frac{d^4k}{(2\pi)^4}\frac{\bar{u}(p')\left[i\frac{g}{\cos\theta_W}\gamma^\beta g_{VA}\right]\left[i(-\not{k} + m_\mu)\right]\left[-ie\gamma^\lambda\right]\left[i(-\not{q}-\not{k}+m_\mu)\right]\left[i\frac{g}{\cos\theta_W}\gamma^\alpha g_{VA}\right]u(p)}{(k^2 - m_\mu^2)\left((q+k)^2 - m_\mu^2\right)\left((k+p')^2 - m_Z^2\right)}\;\frac{-i\,g_{\alpha\beta}}{1}$$

, where $g_{VA} = g_V^f - g_A^f\gamma_5$, with $g_V = \left(\frac{1}{2}T_f^3 - Q_f\sin^2\theta_W\right)$ and $g_A^f = \frac{1}{2}T_f^3$, with $T_f^3 = -\frac{1}{2}$ and $Q_f = -1$ for muon

$\Rightarrow$ $g_V = \frac{1}{4}\left(4\sin^2\theta_W - 1 + \gamma^5\right)$, let's write $\cos\theta_W \equiv C_W$ and $\sin\theta_W \equiv S_W$ for easier notation

taking the limit $M_Z^2 \gg m_\mu^2$ removes all the m_μ^2 from the denominator

$$= -e\frac{g^2}{16C_W^2}\int\frac{d^4k}{(2\pi)^4}\frac{\bar{u}(p')\left[\gamma^\beta(4S_W^2 - 1 + \gamma^5)(-\not{k}+m_\mu)\gamma^\lambda(-\not{q}-\not{k}+m_\mu)\gamma_\beta(4S_W^2 - 1 + \gamma^5)^2\right]u(p)}{k^2\,(q+k)^2\left[(k+p')^2 - m_Z^2\right]}$$

The parity conservation means that we only drop terms linear in γ^5, so the $(\gamma^5)^2 = \mathbb{1}_4$ term survies !! The term with two γ^5 gives us (using $\{\gamma^5, \gamma^\mu\} = 0$) :

$$\gamma^5(-\not{k}+m_\mu)\gamma^\lambda(-\not{q}-\not{k}+m_\mu)\gamma_\beta\gamma^5 = -(\not{k}+m_\mu)\gamma^5\gamma^\lambda(-\not{q}-\not{k}+m_\mu)\gamma^5\gamma_\beta$$

$$= (\not{k}+m_\mu)\gamma^\lambda\gamma^5\gamma^5(\not{q}+\not{k}+m_\mu)\gamma_\beta = (\not{k}+m_\mu)\gamma^\lambda(\not{q}+\not{k}+m_\mu)$$

$$\gamma^5(-\not k + m_\mu)\gamma^\lambda(-\not q - \not k + m_\mu)\gamma_\beta\gamma^5 = -(\not k + m_\mu)\gamma^5\gamma^\lambda(-\not q - \not k + m_\mu)\gamma^5\gamma_\beta$$
$$= (\not k + m_\mu)\gamma^\lambda\gamma^5\gamma^5(\not q + \not k + m_\mu)\gamma_\beta = (\not k + m_\mu)\gamma^\lambda(\not q + \not k + m_\mu)$$

This gives us the Numerator N^λ:

$$N^\lambda = \bar u(p')\left[\gamma^\beta(\not k + m_\mu)\gamma^\lambda(\not q + \not k + m_\mu)\gamma_\beta + (4s_W^2 - 1)^2\gamma^\beta(\not k - m_\mu)\gamma^\lambda(\not q + \not k - m_\mu)\gamma_\beta\right]u(p)$$

Let's now use Feynman parametrization: the two comes from formulae of Feyn. parametrization for 3 terms

$$= -e\frac{g^2}{16 c_W^2}\int_0^1 dx\int_0^{1-x} dy\int\frac{d^4k}{(2\pi)^4}\frac{2\cdot N^\lambda}{\left[x[(k+p')^2 - m_Z^2] + y(q+k)^2 + (1-x-y)k^2\right]^3}$$

$$D \equiv x[(k+p')^2 - m_Z^2] + y(q+k)^2 + (1-x-y)k^2 = x[k^2 + 2p'k + p'^2 - m_Z^2] + y[q^2 + 2qk + k^2] + (1-x-y)k^2$$

$$D = k^2 + 2[xp' + yq]k + xp'^2 + yq^2 - xm_Z^2 \qquad \| \text{ let's do a shift } \ell = k + [xp' + yq]$$

$$= \ell^2 - [xp' + yq]^2 + xp'^2 + yq^2 - xm_Z^2 = \ell^2 - [x^2p'^2 + y^2q^2 + 2xyp'q] + xp'^2 + yq^2 - m_Z^2 x$$

$$= \ell^2 - [x(x-1)p'^2 + y(y-1)q^2 + 2xyp'q] - m_Z^2 x \qquad \| \ \Delta \equiv x(x-1)p'^2 + y(y-1)q^2 + 2xyp'q + xm_Z^2$$

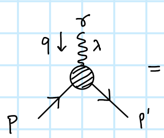
$$= \ell^2 - \Delta$$

$\Rightarrow$ 

$$= -e\frac{g^2}{16 C_W^2}\int dx dy\int\frac{d^4 l}{(2\pi)^4}\frac{2N^\lambda}{(\ell^2 - \Delta)^3},$$

with $N^\lambda = \bar u(p')\left[\gamma^\beta(\not k + m_\mu)\gamma^\lambda(\not q + \not k + m_\mu)\gamma_\beta + (4s_W^2 - 1)^2\gamma^\beta(\not k - m_\mu)\gamma^\lambda(\not q + \not k - m_\mu)\gamma_\beta\right]u(p)$

Let's now simplify this term to extract the contribution to $F_2(q^2)$ of following diagram:



$$= \bar u(p')\, ie\Gamma^\lambda(p,p')\, u(p) \quad \text{, where} \quad ie\Gamma^\lambda = ie\left(\gamma^\lambda F_1(q^2) + \frac{i}{2m}\sigma^{\lambda\beta}q_\beta F_2(q^2)\right)$$

This means that we can drop terms of form Scalar$\cdot\gamma^\lambda$ as they come by since they contribute to $F_1(q^2)$. We can also drop terms linear in ℓ since odd powers of ℓ vanish during integration due to symmetry. This means we can drop all ℓ-terms in N^λ since ℓ^2-terms are proportional to $\gamma^\lambda\cdot$Scalar.

Let's simplify:

$$N^\lambda = \bar u(p')\left[\gamma^\beta(\not k + m_\mu)\gamma^\lambda(\not q + \not k + m_\mu)\gamma_\beta + (4s_W^2 - 1)^2\gamma^\beta(\not k - m_\mu)\gamma^\lambda(\not q + \not k - m_\mu)\gamma_\beta\right]u(p)$$

$$N^\lambda = \bar u(p')\Big\{\gamma^\beta\not k\gamma^\lambda(\not q + \not k)\gamma_\beta + m_\mu\left[\gamma^\beta\not k\gamma^\lambda\gamma_\beta + \gamma^\beta\gamma^\lambda(\not q + \not k)\gamma_\beta\right] + m_\mu^2\underbrace{\gamma^\beta\gamma^\lambda\gamma^\beta}_{=-2\gamma^\lambda \Rightarrow \text{drop}}$$
$$+ (4s_W^2 - 1)^2\Big[\gamma^\beta\not k\gamma^\lambda(\not q + \not k)\gamma_\beta - m_\mu\left[\gamma^\beta\gamma^\lambda(\not q + \not k)\gamma_\beta + \gamma^\beta\not k\gamma^\lambda\gamma_\beta\right] + \underbrace{m_\mu^2\gamma_\beta\gamma^\lambda\gamma_\beta}_{\text{drop}}\Big]\Big\}u(p)$$

Let's now use some Numerator algebra from Peskin appendix A.3:

$$\gamma^\beta\gamma^\alpha\gamma^\lambda\gamma^\delta\gamma_\beta = -2\gamma^\delta\gamma^\lambda\gamma^\alpha \Rightarrow \gamma^\beta\not k\gamma^\lambda(\not q + \not k)\gamma_\beta = -2(\not q + \not k)\gamma^\lambda\not k$$

$$\gamma^\beta\gamma^\alpha\gamma^\lambda\gamma_\beta = 4g^{\alpha\lambda} \Rightarrow \gamma^\beta\not k\gamma^\lambda\gamma_\beta = 4k^\lambda, \quad \gamma^\beta\gamma^\lambda(\not q + \not k)\gamma_\beta = q^\lambda + k^\lambda$$

$$\Rightarrow \quad N^\lambda = \bar{u}(p')\left[-2(\not{q}+\not{k})\gamma^\lambda\not{k}\,[1+(4s_W^2-1)^2] + 4m_\mu[k^\lambda + q^\lambda + k^\lambda](1-(4s_W^2-1)^2)\right]u(p)$$

$$N^\lambda = -2[1+(4s_W^2-1)^2]\,\bar{u}(p')[(\not{q}+\not{k})\gamma^\lambda\not{k}]u(p) + 4m_\mu(1-(4s_W^2-1)^2)\,\bar{u}(p)[q^\lambda + 2k^\lambda]u(p)$$

Let's simplify term by term :

$$\bar{u}(p')[(\not{q}+\not{k})\gamma^\lambda\not{k}]u(p) \qquad \| \; k = l - xp' - yq \qquad q = p'-p \Rightarrow p' = p+q$$

$$= \bar{u}(p')[(\not{q}+\not{l}-x\not{p}'-y\not{q})\gamma^\lambda(\not{l}-x\not{p}'-y\not{q})]u(p) \qquad \| \text{ As mentioned earlier, we can drop } l\text{'s}$$

$$= \bar{u}(p')[[\not{q}(1-y)-xm_\mu]\gamma^\lambda(-(x+y)\not{q}-xm_\mu)]u(p) \qquad \| \text{ Dirac eq. : } \bar{u}(p')\not{p}' = m_\mu\bar{u}(p'), \quad \not{p}\,u(p) = m_\mu u(p)$$

$$= \bar{u}(p')[-(1-y)(x+y)\,\not{q}\gamma^\lambda\not{q} - x(1-y)m_\mu\not{q}\gamma^\lambda + x(x+y)m_\mu\gamma^\lambda\not{q} + \underbrace{x^2m_\mu^2\gamma^\lambda}_{\text{drop}}]\,u(p)$$

$\not{q}\gamma^\lambda\not{q} = (2q^\lambda - \gamma^\lambda\not{q})\not{q} = 2q^\lambda\not{q} - \gamma^\lambda q^2$, which we can also drop, $\gamma^\lambda q^2$ can be dropped

it has the form scalar $\cdot\,\gamma^\lambda$ and $\bar{u}(p')[2q^\lambda\not{q}]u(p)$ can be dropped due to

Ward identity which states $q_\lambda\Gamma^\lambda = 0$. , continuing from previous equality we get

$$= \bar{u}(p')\,xm_\mu[\gamma^\lambda\not{q}\,(x+y) - (1-y)\not{q}\gamma^\lambda]u(p) \qquad \| \text{ Now we again use } \gamma^\lambda\not{q} = 2q^\lambda - \not{q}\gamma^\lambda \text{ and invoke ward identity}$$

$$= \bar{u}(p')\,xm_\mu[\gamma^\lambda\not{q}\,[x+y+1-y]]u(p)$$

$$= \bar{u}(p)[m_\mu x(1+x)\gamma^\lambda\not{q}]u(p)$$

Now let's simplify the other term:

$$\bar{u}(p)[q^\lambda + 2k^\lambda]u(p) = \bar{u}(p')[q^\lambda + 2(l-xp'-yq)^\lambda]u(p) \qquad \| \text{ let's drop } l^\lambda \text{ and } q^\lambda \text{ (Ward identity)}$$

$$= \bar{u}(p')[-2xp'^\lambda]u(p) \qquad \| \; 2p'^\lambda = \{\gamma^\lambda, \not{p}'\}$$

$$= -\bar{u}(p')[x\gamma^\lambda\not{p}' + \underbrace{xm_\mu\gamma^\lambda}_{\text{drop}}]u(p) \qquad \| \; \gamma^\lambda\not{p}'u(p) = \gamma^\lambda(\not{q}+\not{p})u(p) = [\gamma^\lambda\not{q} + \underbrace{\gamma^\lambda m_\mu}_{\text{can be dropped}}]u(p)$$

$$= \bar{u}(p')[-x\gamma^\lambda\not{q}]u(p)$$

With this, N^λ becomes:

$$N^\lambda = \bar{u}(p')[-2(1+(4s_W^2-1)^2)m_\mu x(1+x)\gamma^\lambda\not{q} - 4m_\mu(1-(4s_W^2-1)^2)x\gamma^\lambda\not{q}]u(p)$$

$$= \bar{u}(p')[(4s_W^2-1)^2\underbrace{[-2x(1+x)+4x]}_{4x-2x-2x^2 = 2x(1-x)}m_\mu\gamma^\lambda\not{q} + \underbrace{(-2x(1+x)-4x)}_{-4x-2x-2x^2 = -2x(3+x)}m_\mu\gamma^\lambda\not{q}]u(p)$$

$$N^\lambda = \bar{u}(p')[2x(1-x)(4s_W^2-1)^2 - 2x(3+x)]m_\mu\gamma^\lambda\not{q}\;u(p)$$

Now withe help of ward identity we can write

$$\gamma^\lambda\not{q} = \frac{1}{2}[\gamma^\lambda\not{q} + \gamma^\lambda\not{q}] \overset{\checkmark}{=} \frac{1}{2}[\gamma^\lambda\not{q} - \not{q}\gamma^\lambda] = \frac{1}{2}[\gamma^\lambda, \gamma^\beta]q_\beta$$

(equality since ward identity eliminates q^λ term) equality since ward identity eliminates q^λ term

$$\gamma^\lambda \not{q} = \frac{1}{2}\left[\gamma^\lambda \not{q} + \gamma^\lambda \not{q}\right] \overset{\checkmark}{=} \frac{1}{2}\left[\gamma^\lambda \not{q} - \not{q}\gamma^\lambda\right] = \frac{1}{2}\left[\gamma^\lambda, \gamma^\beta\right] q_\beta$$

Using definition of $\sigma^{\lambda\beta}$, $\sigma^{\lambda\beta} = \frac{i}{2}\left[\gamma^\lambda, \gamma^\beta\right]$, we get

$$\Rightarrow \quad N^\lambda = \left[\, 2x(1-x)(4S_W^2-1)^2 - 2x(3+x)\right] m_\mu\, \bar{u}(p')(-i\sigma^{\lambda\beta}q_\beta)\, u(p)$$

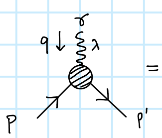
$$N^\lambda = \left[2x(3+x) - 2x(1-x)(4S_W^2-1)^2\right] 2m_\mu^2 \quad \bar{u}(p')\left[i\,\frac{\sigma^{\lambda\beta}}{2m_\mu}q_\beta\right] u(p)$$

$\Rightarrow$ 
$$= -e\,\bar{u}(p')\left[-i\,\frac{\sigma^{\lambda\beta}}{2m_\mu}q_\beta\right]u(p)\,\frac{g^2 2}{16C_W^2}\int_0^1 dx\int_0^{1-x}\left[2x(3+x)-2x(1-x)(4S_W^2-1)^2\right]2m_\mu^2\int\frac{d^4\ell}{(2\pi)^4}\,\frac{1}{(\ell^2-\Delta)^3}$$

We can get the ℓ-integral from peskin eq. (A.44): $\displaystyle\int\frac{d^4\ell}{(2\pi)^4}\,\frac{1}{(\ell^2-\Delta)^3} = \frac{-i}{(4\pi)^2}\,\frac{1}{2}\,\frac{1}{\Delta}$


$$= ie\,\bar{u}(p')\left[-\frac{i\sigma^{\lambda\beta}}{2m_\mu}q_\beta\right]u(p)\,2\,\frac{g^2}{16C_W^2}\,\frac{m_\mu^2}{(4\pi)^2}\int_0^1 dx\int_0^{1-x}dy\,\frac{\left[2x(3+x)-2x(1-x)(4S_W^2-1)^2\right]}{\Delta}$$

Looking at the following diagram,



$$= \bar{u}(p')\, ie\Gamma^\lambda(p,p')\,u(p) \quad \text{, where} \quad ie\Gamma^\lambda = ie\left(\gamma^\lambda F_1(q^2) + \frac{i}{2m}\sigma^{\lambda\beta}q_\beta\, F_2(q^2)\right),$$

we see that we have gotten the correct contribution to $F_2(q^2)$ which is:

$$-\frac{g^2}{8C_W^2}\,\frac{m_\mu^2}{(4\pi)^2}\int_0^1 dx\int_0^{1-x}dy\,\frac{\left[2x(3+x)-2x(1-x)(4S_W^2-1)^2\right]}{\Delta(q^2)} \qquad \Big\|\ \Delta \text{ is function of } q^2$$

The anomalous magnetic moment is $F_2(0)$ so the contribution from Z-boson vertex is

$$a_\mu(Z) = -\frac{g^2}{8C_W^2}\,\frac{m_\mu^2}{(4\pi)^2}\int_0^1 dx\int_0^{1-x}dy\,\frac{\left[2x(3+x)-2x(1-x)(4S_W^2-1)^2\right]}{\Delta(0)}$$

Let's now approximate Δ in the limit $m_\mu \ll m_Z^2$

$$\Delta(q^2) = x(x-1)p'^2 + y(y-1)q^2 + 2xy\,p'q + x m_Z^2$$

$\Big\|$ when $q^2 = 0$ and $p'^2 = p^2 = m_\mu^2 \ll m_Z^2$ only the m_Z^2 term remains

$$\Rightarrow \quad \Delta(0) \simeq x\, m_Z^2$$

$$\Rightarrow \quad a_\mu(Z) = -\frac{g^2}{8C_W^2}\,\frac{m_\mu^2}{16\pi^2}\,\frac{1}{m_Z^2}\int_0^1 dx\int_0^{1-x}dy\quad 2(3+x) - 2(1-x)\left[4S_W^2-1\right]^2$$

$$= -\frac{G_F m_\mu^2}{16\pi^2}\,\frac{1}{\sqrt{2}}\int_0^1 dx\ 2(3+x)(1-x) - 2(1-x)^2\left[4S_W^2-1\right]^2$$

$$= +\frac{G_F m_\mu^2}{8\pi^2}\,\frac{1}{\sqrt{2}}\,\frac{1}{3}\left[(4S_W^2-1)^2 - 5\right]$$

$$= +\frac{G_F m_\mu^2}{8\pi^2}\,\frac{1}{\sqrt{2}}\,\frac{1}{3}\left[16S_W^4 - 8S_W^2 + 1 - 5\right]$$

$\Big\|$ $m_Z^2 = \dfrac{m_W^2}{C_W^2}$

$\dfrac{g^2}{8m_W^2} = G_F\,\dfrac{1}{\sqrt{2}}$

Integral gives $-\frac{2}{3}\left[(4S_W^2-1)^2 - 5\right]$

$$\Rightarrow \quad a_\mu(Z) = -\frac{G_F m_\mu^2}{8\pi^2\sqrt{2}}\,\frac{1}{3}\left(4 + 8\sin^2\theta_W - 16\sin^4\theta_W\right) \quad \square$$

$$\Rightarrow \quad a_\mu(Z) = -\frac{G_F\, m_\mu^2}{8\pi^2\sqrt{2}}\,\frac{1}{3}\left(4 + 8\sin^2\theta_W - 16\sin^4\theta_W\right) \qquad \square$$

We have derived the correct contribution!!

Thanks for extending the deadline!!